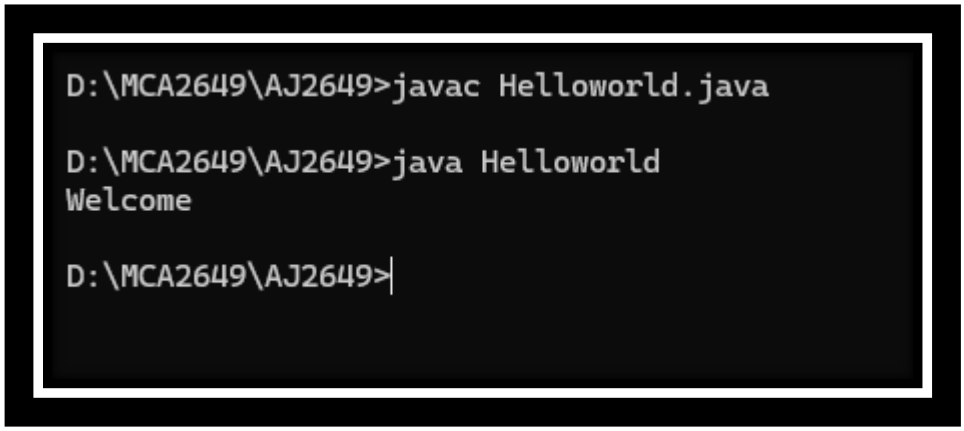
Assignment No	1
Title	Learning and Understanding :- Factorial of a number & Hello World
Objective	1A).Hello World program without creating an object 2A). Factorial program without creating an object 3B). Hello World program with creating an object 4B). Factorial program with creating an object
Roll No	MCA2649

Source Code:-

1A).Hello World program without creating an object

```
class Helloworld
{
	public static void main(String []args)
	{
		String a = "Welcome";
		String result = helloWorld(a);
		System.out.println(result);
	}
	public static String helloWorld(String s)
	{
		return s;
	}
}
```

Output:-



2A). Factorial program without creating an object

```
import java.util.*;
class FactorialApp
{
        public static void main(String [] args)
        {

                //Scanner Class
                Scanner sc = new Scanner(System.in);
                System.out.println("Enter the number :");
                int number = sc.nextInt();

                //local variable
                int fact = 1;
```

```
            //logic
            while(number>=1)
            {
                fact = fact*number;
                number = number-1;
            }

            //print
            System.out.println("Factorial Number :" + fact);
        }
}
```

Output:-

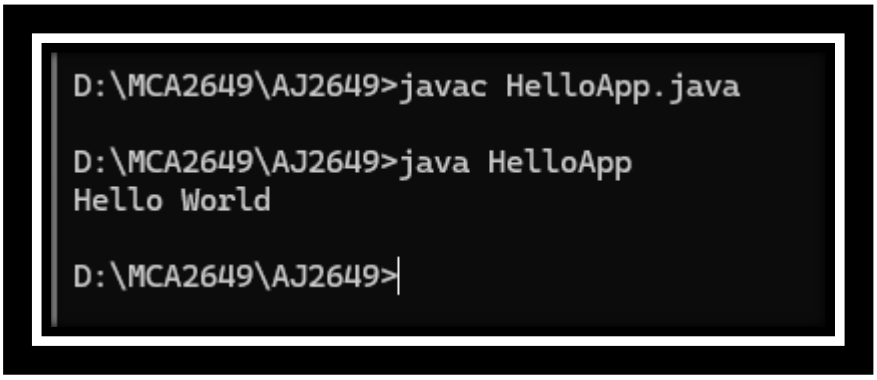


3B). Hello World program with creating an object

```
class HelloApp
{
	public static void main(String args[])
	{
		HelloApp obj = new HelloApp();
		obj.helloAppMethod();

	}
	public void helloAppMethod()
	{
		System.out.println("Hello World");
	}
}
```

Output:-



4B). Factorial program with creating an object

```
class FactorialImpl
```

```
{
        //Instance Declaration
        int number;
        int fact = 1;

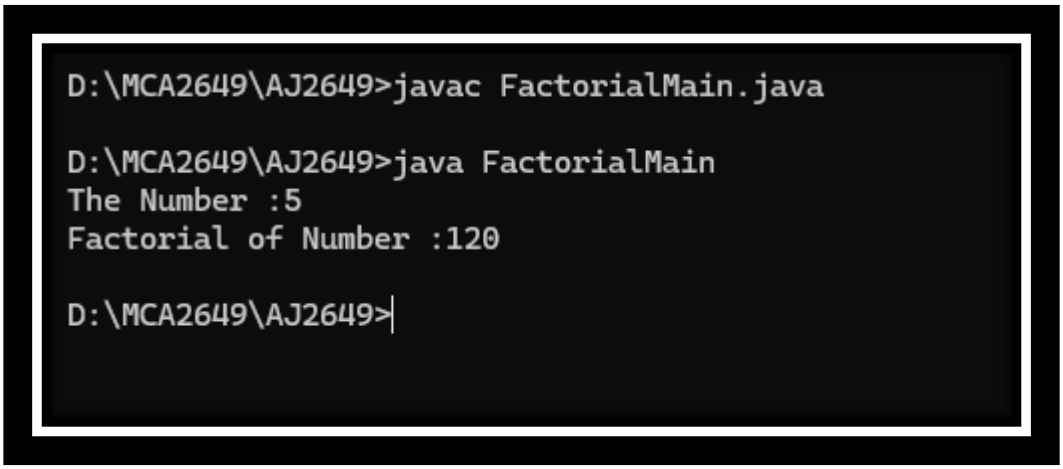
        //Getter/Setter
        public void set(int number)
        {
                this.number = number;
        }
        int get()
        {
                return number;
        }
        //factorial logic
        int factorialMethod()
        {
                while(number>=1)
                {
                        fact = fact*number;
                        number = number-1;
                }
                return fact;
        }
```

```
        //print
        void print()
        {
                System.out.println("The Number :" + number);
        }
}

class FactorialMain
{
        public static void main(String args[])
        {
                FactorialImpl obj = new FactorialImpl();
                obj.set(5);
                obj.get();
                obj.print();

                int res = obj.factorialMethod();
                System.out.println("Factorial of Number :" + res);
        }
}
```

Output:-


D:\MCA2649\AJ2649>javac FactorialMain.java
D:\MCA2649\AJ2649>java FactorialMain
The Number :5
Factorial of Number :120
D:\MCA2649\AJ2649>